



Dynamic simulation of spur gear with tooth root crack propagating along tooth width and crack depth

Zaigang Chen, Yimin Shao *

State Key Laboratory of Mechanical Transmission, Chongqing University, Chongqing 400030, China

ARTICLE INFO

Article history:

Received 11 April 2011
Received in revised form 26 June 2011
Accepted 8 July 2011
Available online 22 July 2011

Keywords:

Tooth crack
Propagation
Gear mesh stiffness
Sidebands

ABSTRACT

Gear tooth crack will cause changes in vibration characteristics of gear system, based on which, operating condition of the gear system is always monitored to prevent a presence of serious damage. However, it is also a unsolved puzzle to establish the relationship between tooth crack propagation and vibration features during gear operating process. In this study, an analytical model is proposed to investigate the effect of gear tooth crack on the gear mesh stiffness. Both the tooth crack propagations along tooth width and crack depth are incorporated in this model to simulate gear tooth root crack, especially when it is at very early stage. With this analytical formulation, the mesh stiffness of a spur gear pair with different crack length and depth can be obtained. Afterwards, the effects of gear tooth root crack size on the gear dynamics are simulated and the corresponding changes in statistical indicators – RMS and kurtosis are investigated. The results show that both RMS and kurtosis increase with the growth of tooth crack size for propagation whatever along tooth width and crack length. Frequency spectrum analysis is also carried out to examine the effects of tooth crack. The results show that sidebands caused by the tooth crack are more sensitive than the mesh frequency and its harmonics. The developed analytical model can predict the change of gear mesh stiffness with presence of a gear tooth crack and the corresponding dynamic responses could supply some guidance to the gear condition monitoring and fault diagnosis, especially for the gear tooth crack at early stage.

© 2011 Elsevier Ltd. All rights reserved.

0. Introduction

Gearboxes are the most important mechanisms in industrial machinery, automotive applications, and our daily lives to transmit power and produce high rotational speed changes and/or change the direction of motion. And due to their growing applications, gearbox health monitoring and early fault detection have been under intensive investigation [1,2].

As is known, gear tooth failure can cause removal and/or plastic deformations on the contacting tooth surfaces or even presence of fatigue crack. And the severity of tooth damage is usually assessed by the reduction of the stiffness [3,4]. There has been a lot of work carried out to investigate gear tooth stiffness with and/or without tooth faults. Finite element models (FEA) [4–8] and analytical methods are the widely used approaches to fulfill the stiffness modeling and calculation. However, FEA models for the tooth stiffness calculation need mesh refinements and are computationally expensive. On the other hand, analytical methods show good results with less computation time compared with FEA models [4,9,10].

Gear mesh stiffness without defects was computed analytically by Weber [9], Cornell [10], while a digitization approach was used by Kasuba and Evans [11]. Yang and Lin [12] used the so-called potential energy method to calculate the total mesh stiffness of a gear pair versus gear rotational position. And their model was further refined by Tian [13] and Wu et al. [1] by

* Corresponding author. Tel.: +86 (0)23 65112520; fax: +86 (0)23 65106195.

E-mail address: yimshao@cqu.edu.cn (Y. Shao).

taking the shear mesh stiffness into consideration. But, they did not take the fillet-foundation deflections into account yet. All the former research papers on the analytically modeling of gear tooth crack assumed that the tooth crack was through the whole tooth width with a constant crack depth. In this paper, the crack propagating along tooth width and crack depth is modeled and its influences on gear mesh stiffness and dynamic responses are investigated.

Lewicki [14–16] did many works on the tooth crack propagation path and gained some useful conclusions. He noted that crack propagation paths depend on the backup ratio which is defined as the ratio of rim thickness to tooth height and it tends to be smooth, continuous, and rather straight with only a slight curvature. For the gear with high backup ratio, gear tooth root crack would propagate through the tooth along tooth width. While for those with low backup ratio, the crack would go through the rim. The initial crack angle is also a determinative factor for the propagation. In the case of low initial crack angle, the propagation is through the rim even with high backup ratio. In this paper, the crack along its depth direction is also assumed to be straight but with a non-uniform distribution along tooth width, which is more realistic and different from those proposed by Wu et al. [1] and Chaari et al. [4]. This model makes it possible to check the effectiveness of algorithms in fault diagnosis and condition monitoring, especially for the crack at early stage.

The earliest review papers on the numerical modeling of spur gear dynamics are made by Özgüven and Houser in 1988 [17] and by Parey and Tandon in 2003 [18]. But, there are few reviews on the gear tooth defects. Randall [19,20] reported the advantages of simulating faults in machine such as producing sufficient representative signals to train automated fault recognition algorithms, generating signals for faults with different sizes and locations in order to test and compare diagnostic algorithms and being very helpful in gaining a physical understanding of the complex (often nonlinear) interactions. In his work, gear tooth crack and spall were involved. Chaari et al. [3] stated that dynamic responses of the transmission is closely related to the time varying gear mesh stiffness, and higher vibration and acoustic emission level is noticed when the mesh stiffness is reduced due to some gear tooth faults. Modelling of the gear transmission failure would help to analyze the change in dynamic characteristics which can be a suitable tool for maintenance teams to diagnose such failures. Li et al. [21] developed an embedded modelling approach for identifying gear meshing stiffness from measured gear angular displacement or transmission error and then using an embedded–dynamic–fracture model to predict gear fatigue crack propagation [22]. Endo et al. [23,24] presented a technique to differentially diagnose two types of localized gear tooth faults: a spall and a crack in the gear tooth fillet region by simulation and experiment. These researches may provide useful information for fault detection. Vibration-based time domain, frequency domain, and time–frequency domain analyses are the most powerful tools available for fault detection of rotating machinery [1]. Traditional techniques based on statistical measurements and vibration spectrum analysis are applied to analyze the effects of tooth crack on the gear vibrations in this paper.

Based on the discussions above, the published papers on the gear tooth root crack are always under the assumption that the crack is through the whole tooth width with the same depth. In reality, the crack is not always through the whole tooth width but from an initial position where a stress concentration is observed. In the present paper, the tooth crack propagating along tooth width is formulated and an example is also simulated and analyzed.

The main objective of this paper is to develop an analytical model of mesh stiffness for spur gear pair to predict the tooth root crack which propagates along both tooth width and crack depth. Thus, it can make up the shortage of previous research that they are limited to the assumption that the tooth crack is through the whole tooth width with a constant depth. And the developed model is validated by comparisons with FEA results in Ref. [4]. With the developed model, the effects of gear tooth root crack with different length and depth on the mesh stiffness are investigated. After obtaining the gear mesh stiffness with tooth root crack, a six-degree-of-freedom spur gear system is established to investigate the influence of gear tooth root crack on its dynamic responses. The statistical indicators—RMS and kurtosis, which are widely used in fault diagnosis and condition monitoring, are applied to expose the influence of gear tooth crack. Analysis of the dynamic responses in time – and frequency-domain are also carried out so that the effects of tooth crack on the gear dynamic characteristics are obtained quantitatively. Thus, the developed analytical model can be used to assess the impact of gear tooth crack with different sizes on the gear mesh stiffness and the corresponding dynamic responses, which is useful in the gear condition monitoring and fault diagnosis, especially for the gear tooth crack at early stage.

This paper is organized as follows: reviews on gear tooth fault in the published papers and other documentations are given in introduction. The first segment introduces Wu et al. [1] tooth crack model and extend it to include the effect of fillet-foundation deformation, based on which, the crack model to simulate the propagation along tooth width is also developed in this part. And the investigations on the effects of gear tooth root crack on gear mesh stiffness are carried on in the second part. Then, after incorporating the gear mesh stiffness with different sizes into gear dynamic model, the effects of the gear tooth root crack on dynamic responses of gear system are analyzed in the third section which is followed by conclusions.

1. Modeling of gear mesh stiffness with a crack at tooth root

For the analytical studies on the tooth crack of spur gears, there has been many works published. Nearly all of them assumed the crack is through the tooth width with a constant crack depth. Few work focus their attentions on the gear tooth propagating along tooth width, namely the crack depth varies along tooth width. A gear mesh stiffness calculation model with consideration of tooth crack propagating along tooth width is proposed in Section 1.2 which is based on the existing analytical model and method reviewed in Section 1.1 which assumes the tooth crack was though the tooth width with a constant depth.

1.1. Mesh stiffness calculation with crack through tooth width

Deflections of a spur gear tooth can be determined by considering it as a non-uniform cantilever beam with an effective length d displayed in Fig. 1. Here, the crack is assumed to go through the whole tooth width $[1,4,13]$ W with a constant depth q_0 and a crack inclination angle α_c .

The bending, shear and axial compressive energy stored in a tooth can be represented by [12,13],

$$U_b = \frac{F^2}{2K_b}, \quad U_s = \frac{F^2}{2K_s}, \quad U_a = \frac{F^2}{2K_a} \quad (1-a, b, c)$$

where K_b , K_s , K_a are the bending, shear and axial compressive stiffness in the same direction under the action of the force F .

Based on the beam theory, the potential energy stored in a meshing gear tooth can be calculated by [1,12,13],

$$U_b = \int_0^d \frac{M^2}{2EI_x} dx, \quad U_s = \int_0^d \frac{1.2F_b^2}{2GA_x} dx, \quad U_a = \int_0^d \frac{F_a^2}{2EI_x} dx \quad (2-a, b, c)$$

where U_b , U_s , U_a are the potential energy stored in the bending, shear and axial compressive deformations, respectively under the action of the mesh force F . And F_b , F_a and M are calculated by

$$F_b = F \cos \alpha_1, \quad F_a = F \sin \alpha_1, \quad M = F_b \cdot x - F_a \cdot h \quad (3-a, b, c)$$

Based on Eqs. (1)–(3), the bending stiffness K_b can be obtained as,

$$\frac{1}{K_b} = \int_0^d \frac{(x \cos \alpha_1 - h \sin \alpha_1)^2}{EI_x} dx \quad (4)$$

Shear stiffness K_s is calculated by,

$$\frac{1}{K_s} = \int_0^d \frac{1.2 \cos^2 \alpha_1}{GA_x} dx \quad (5)$$

Axial compressive stiffness K_a is,

$$\frac{1}{K_a} = \int_0^d \frac{\sin^2 \alpha_1}{EA_x} dx \quad (6)$$

In the formulas (2)–(6), h , x , dx , α_1 , d are shown in Fig. 1. E is the Young modulus. G represents the shear modulus. I_x and A_x represent the area moment of inertia and area of the section where the distance between the section and the acting point of the applied force is x , and they can be obtained by

$$I_x = \begin{cases} \frac{1}{12} (h_x + h_q)^3 W, & h_x \leq h_q \\ \frac{1}{12} (h_x + h_q)^3 W, & h_x > h_q \end{cases} \quad (7)$$

$$A_x = \begin{cases} (h_x + h_q) W, & h_x \leq h_q \\ (h_x + h_q) W, & h_x > h_q \end{cases} \quad (8)$$

$$G = \frac{E}{2(1 + \nu)} \quad (9)$$

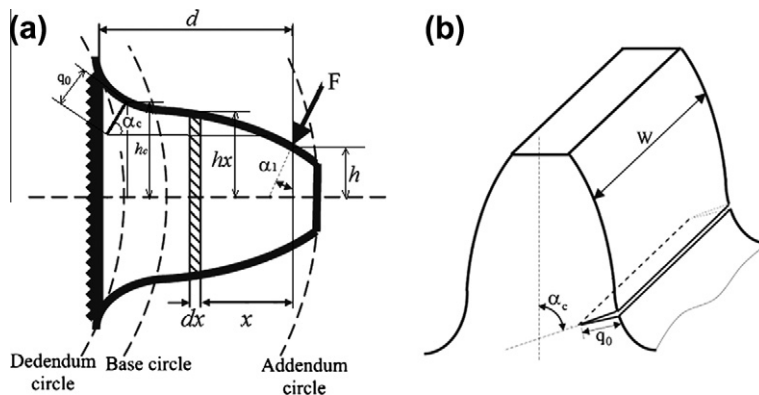


Fig. 1. Model of the spur gear tooth as a non-uniform cantilever beam with a crack at tooth root.

where ν is the Poisson ratio. $h_q = h_c - q_0 \sin \alpha_c$, q_0 and α_c are the depth and the inclination angle of the crack, respectively. h_x represents the distance between a point and the tooth's central line and this point lies on the tooth profile curve where the horizontal distance from the tooth's root is equal to d minus x .

From the results derived by Yang and Sun [25], the stiffness of Hertzian contact of two meshing teeth is constant along the entire line of action. It is independent of the contact position and the interpenetration depth between meshing teeth. The Hertzian contact stiffness K_h is given by

$$\frac{1}{K_h} = \frac{4(1 - \nu^2)}{\pi EW} \quad (10)$$

Besides the tooth deformation, the fillet-foundation deflection also influences the stiffness of gear tooth. Sainsot et al. [26] derived the fillet-foundation deflection of the gear based on the theory of Muskhelishvili [27]. And then, they applied it to circular elastic rings to derive an analytical formula reflecting the gear body-induced tooth deflections by assuming linear and constant stress variations at root circle. It can be calculated as [3,4,7,26],

$$\delta_f = \frac{F \cos^2 \alpha_m}{WE} \left\{ L^* \left(\frac{u_f}{S_f} \right)^2 + M^* \left(\frac{u_f}{S_f} \right) + P^* (1 + Q^* \tan^2 \alpha_m) \right\} \quad (11)$$

where W is the tooth width. u_f and S_f are given in Fig. 2. The coefficients L^* , M^* , P^* , Q^* can be approached by polynomial functions [26]:

$$X_i^*(h_{fi}, \theta_f) = A_i/\theta_f^2 + B_i h_{fi}^2 + C_i h_{fi}/\theta_f + D_i/\theta_f + E_i h_{fi} + F_i \quad (12)$$

X_i^* denotes the coefficients L^* , M^* , P^* and Q^* . $h_{fi} = r_f/r_{int}$, r_f , r_{int} and θ_f are defined in Fig. 2, the values of A_i , B_i , C_i , D_i , E_i and F_i are given in Table 1.

The stiffness with consideration of gear fillet-foundation deflection can be obtained by

$$\frac{1}{K_f} = \frac{\delta_f}{F} \quad (13)$$

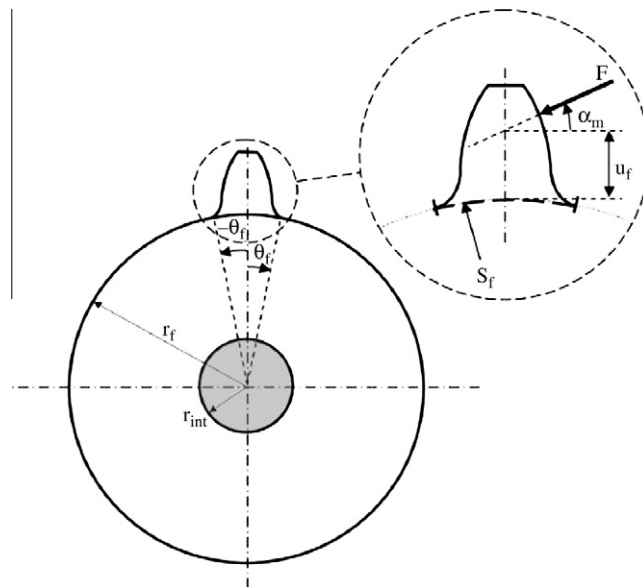


Fig. 2. Geometrical parameters for the fillet-foundation deflection [3,4].

Table 1

Values of the coefficients of Eq. (12) [22].

	A_i	B_i	C_i	D_i	E_i	F_i
$L^*(h_{fi}, \theta_f)$	-5.574×10^{-5}	-1.9986×10^{-3}	-2.3015×10^{-4}	4.7702×10^{-3}	0.0271	6.8045
$M^*(h_{fi}, \theta_f)$	60.111×10^{-5}	28.100×10^{-3}	-83.431×10^{-4}	-9.9256×10^{-3}	0.1624	0.9086
$P^*(h_{fi}, \theta_f)$	-50.952×10^{-5}	185.50×10^{-3}	0.0538×10^{-4}	53.3×10^{-3}	0.2895	0.9236
$Q^*(h_{fi}, \theta_f)$	-6.2042×10^{-5}	9.0889×10^{-3}	-4.0964×10^{-4}	7.8297×10^{-3}	-0.1472	0.6904

The total equivalent mesh stiffness of one tooth pair in mesh can be obtained by

$$K_e = 1 / \left(\frac{1}{K_{b1}} + \frac{1}{K_{s1}} + \frac{1}{K_{a1}} + \frac{1}{K_{f1}} + \frac{1}{K_{b2}} + \frac{1}{K_{s2}} + \frac{1}{K_{a2}} + \frac{1}{K_{f2}} + \frac{1}{K_h} \right) \quad (14)$$

Here, the subscripts 1, 2 mean the pinion and gear, respectively.

1.2. Mesh stiffness model with crack propagating along tooth width

At very early stage of crack propagation, it usually starts from some local positions where the stress concentrations are observed. In order to investigate the effect of a tooth root crack propagating along tooth width non-uniformly, a mesh stiffness model is developed by dividing a gear tooth into some independent thin pieces like that shown in Fig. 3b, so that the crack length along tooth width for each piece can be regarded as a constant which is reasonable when dx is small. Stiffness of each piece is denoted as $K_t(x)$ and can be calculated with taking tooth bending, shear and axial compress into account based on Eqs. (4)–(6). $K_t(x)$ can be obtained by

$$K_t(x) = 1 / \left(\frac{1}{K_b(x)} + \frac{1}{K_s(x)} + \frac{1}{K_a(x)} \right) \quad (15)$$

$K_b(x)$, $K_s(x)$, $K_a(x)$ are the stiffness under consideration of the effect of tooth bending, shear and axial compress for one piece of tooth. x is the distance between the piece of tooth and one end of the tooth which is shown in Fig. 3a and b.

Then, the stiffness of the whole tooth can be obtained by integration of $K_t(x)$ along tooth width,

$$K_t = \int_0^W K_t(x) \quad (16)$$

With the effect of fillet-foundation deflection and Hertzian contact, the equivalent mesh stiffness can be calculated by

$$K_e = 1 / \left(\frac{1}{K_{t1}} + \frac{1}{K_{f1}} + \frac{1}{K_{t2}} + \frac{1}{K_{f2}} + \frac{1}{K_h} \right) \quad (17)$$

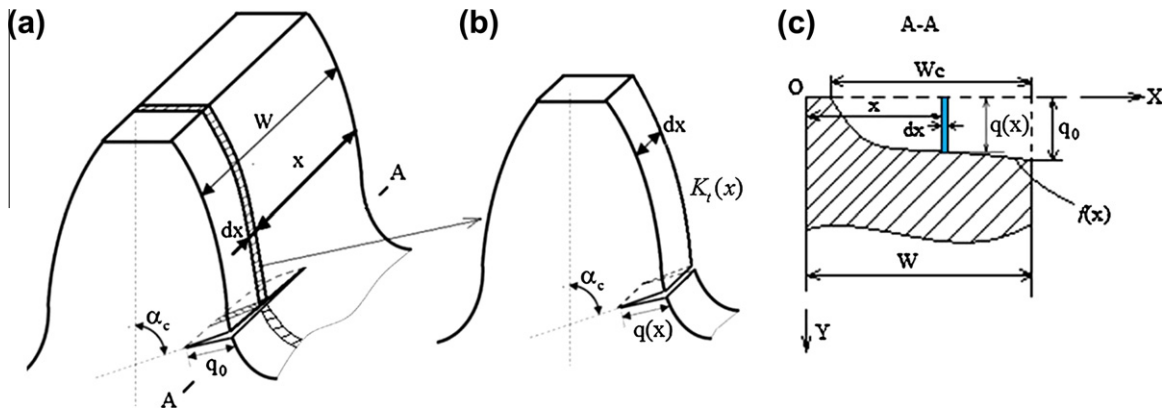


Fig. 3. Crack model at gear tooth root.

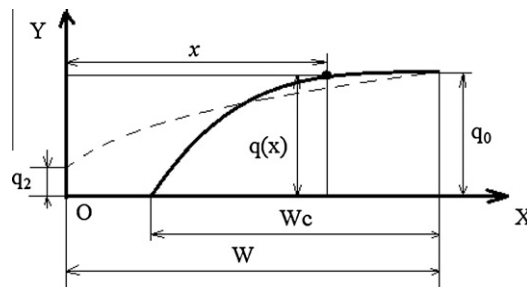


Fig. 4. Crack depth along tooth width.

Although the proposed mesh stiffness model can also be suitable to a spatial gear tooth crack, this study assumes that the crack propagation is in plane shown as A–A in Fig. 3a and c with different crack depth along tooth width for simplicity. The crack depth along tooth width can be described as a function of x in the coordinate system XOY in Figs. 3c and 4. That means

$$qx = f(x) \quad (18)$$

Further, the crack depth is assumed to distribute along tooth width as a parabolic function as shown in Fig. 4. When the crack length W_c is less than tooth width W , the crack curve is denoted by the solid curve in Fig. 4. While the crack propagates through the whole tooth width, the crack curve is described as the dashed curve which propagates along crack depth q_2 .

For the solid curve,

$$\begin{cases} q(x) = q_0 \sqrt{\frac{x+W_c-W}{W_c}}, & x \in [W - W_c, W] \\ q(x) = 0, & x \in [0, W - W_c] \end{cases} \quad (19)$$

For the dashed curve,

$$q(x) = \sqrt{\frac{q_0^2 - q_2^2}{W}x + q_2^2} \quad (20)$$

Parameters of the pinion-wheel set

2. Investigation on effect of tooth crack on gear mesh stiffness

Based on the developed analytical model in this paper, effects of the tooth crack size, namely crack length along tooth width and crack depth, on the mesh stiffness of gear pair are able to be calculated and investigated. The main parameters used in this paper are from Ref. [4] and shown in Table 2.

2.1. Comparisons with FEA results in Ref. [4]

In Ref. [4], healthy case and two crack cases were analyzed. They are:

Crack No. 1: $p_c = 0.3$ mm, $\alpha_c = 33^\circ$, position: beginning at the root circle (in this study, just let $q_0 = q_2 = 0.3$ mm, $\alpha_c = 33^\circ$ to set the same case).

Crack No. 2: $p_c = 0.66$ mm, $\alpha_c = 70^\circ$, position: beginning at the root circle (in this study, just let $q_0 = q_2 = 0.66$ mm, $\alpha_c = 70^\circ$ to set the same case).

The comparisons made in Table 3 and Fig. 5 between the mesh stiffness results from Ref. [4] by FEA and that from the analytical model developed in this study validate the developed model in this paper.

2.2. Effect of tooth root crack propagating along tooth width on gear mesh stiffness

When the crack propagates through the whole tooth width, that means W_c is equal to W shown in Fig. 4, it continues to propagate along the depth direction labeled as q_2 . The case where $q_0 = 1$ mm, $\alpha_c = 60^\circ$ is investigated.

Table 2
Parameters of the pinion-wheel set [4].

	Pinion	Gear
Teeth number	30	25
Module (mm)	2	2
Teeth width (mm)	20	20
Contact ratio	1.63	1.63
Rotational speed (rpm)	2000	2400
Pressure angle	20°	20°
Young modulus E (N/mm ²)	2×10^5	2×10^5
Poisson's ratio	0.3	0.3

Table 3
Mesh stiffness comparison of single tooth pair.

Gear tooth condition	FEM results [4]	Results obtained in this paper	Difference (%)
Healthy case	1.58×10^8	1.52×10^8	3.8
Crack No. 1	1.53×10^8	1.47×10^8	3.9
Crack No. 2	1.42×10^8	1.38×10^8	2.8

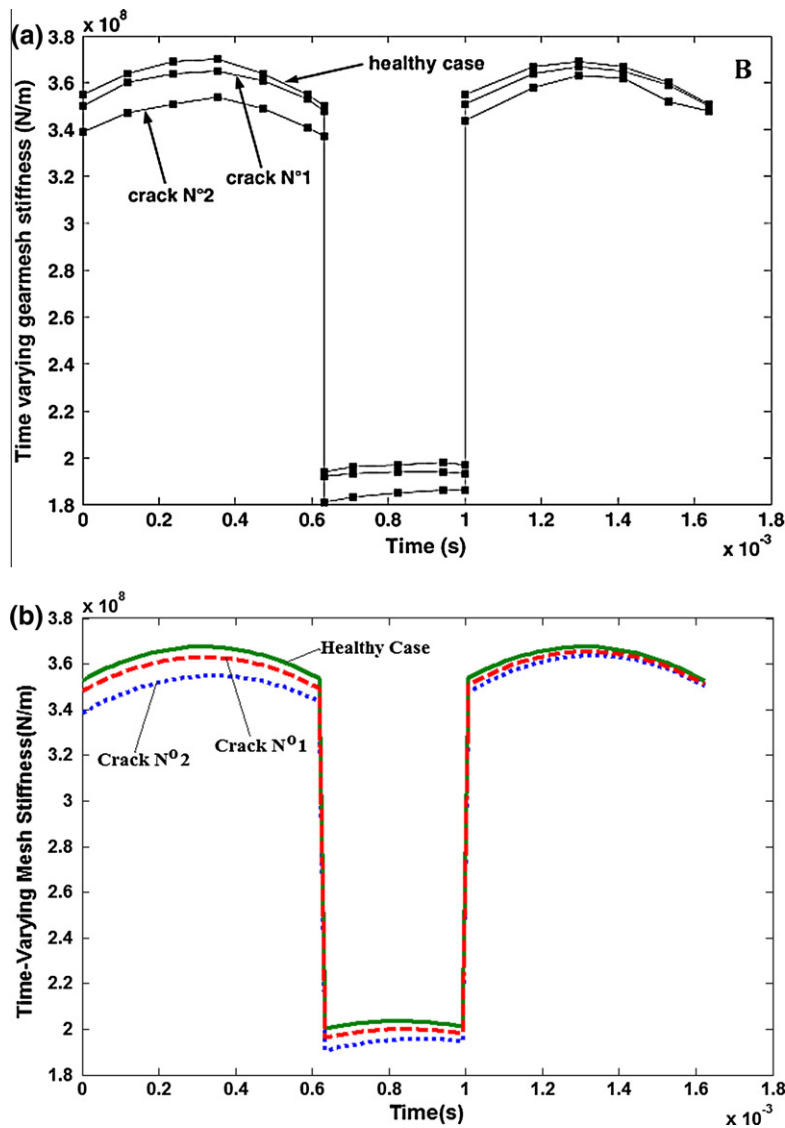


Fig. 5. Time-varying mesh stiffness. (a) Gear mesh stiffness from Ref. [4] by FEA method. (b) Obtained from the model developed in this study.

The mesh stiffness curves of a single tooth pair with different crack lengths are displayed in Fig. 6. These curves are plotted versus the pinion angular position. And the mesh stiffness reflecting the alternative process between single- and double-tooth engagements are shown in Fig. 7. In these two figures, distinct reductions of stiffness are observed when tooth cracks are introduced. The stiffness reduction increases with growth of crack length. The same phenomenon as stated in Ref. [4] can be observed that maximum stiffness reduction for a fixed crack appears where the cracked tooth of pinion is just going to engagement. This is an expected result because of the relative bigger flexibility of the tooth at the addendum circle compared to that at the position on the tooth profile which is closer to base circle. It is noted that the crack length of $(20 + 1)$ mm refers the case when $W_c = W = 20$ mm and $q_2 = q_0 = 1$ mm. In addition, a noticeable result can be obtained that the mesh stiffness reduction due to crack appears to increase more promptly when the crack propagates through the whole tooth width and continues to propagate along crack depth.

2.3. Effect of tooth root crack depth on gear mesh stiffness

The cases where $W_c = 15$ mm, $\alpha_c = 60^\circ$ and q_0 increasing gradually from 0 to 1.2 mm with interval of 0.3 mm are simulated to investigate the effect of tooth crack depth on gear mesh stiffness. The corresponding results for single tooth pair and the alternative process between single- and double-tooth engagements are shown in Figs. 8 and 9, respectively. The same

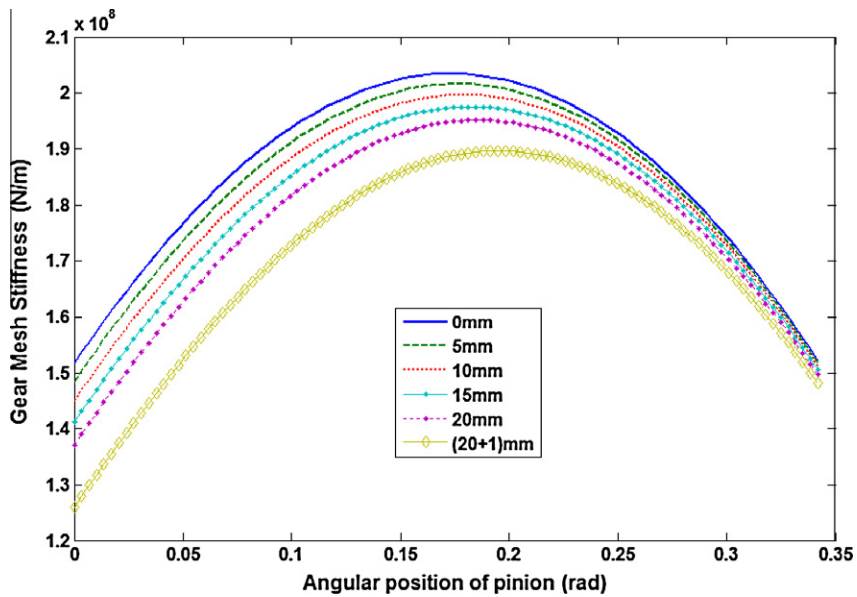


Fig. 6. Mesh stiffness with different crack length along tooth width (single tooth pair).

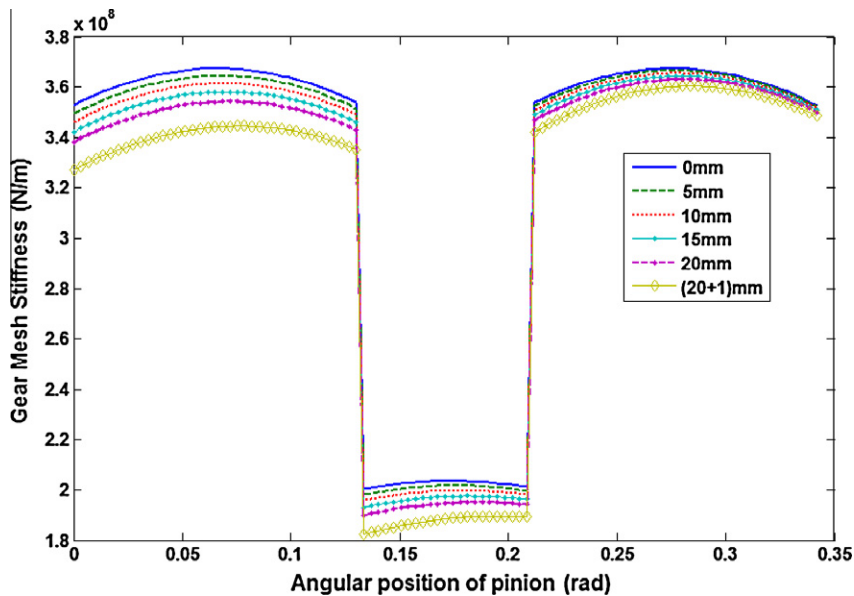


Fig. 7. Mesh stiffness with different crack length along tooth width.

phenomenon is also found that the maximum stiffness reduction during the mesh process for a fixed crack appears at the beginning of the engagement when the cracked tooth of the pinion is getting into engagement.

3. Dynamic simulation of spur gear system with gear tooth crack

3.1. Spur gear system

Based on the mesh stiffness model of gear pair with or/and without a tooth root crack, a dynamic lumped parameter model of a spur gearbox system comprising of six degrees of freedom (DOF) is established. A schematic of the gear dynamic model is shown in Fig. 10 where the y axis is parallel to the line of action (LOA) of the gear pair. T_p/T_g is external/braking torque; Ω_p/Ω_g is nominal operational speeds of pinion/gear; J_p/J_g is inertial moment of pinion/gear; m_p/m_g is mass of pinion/gear;

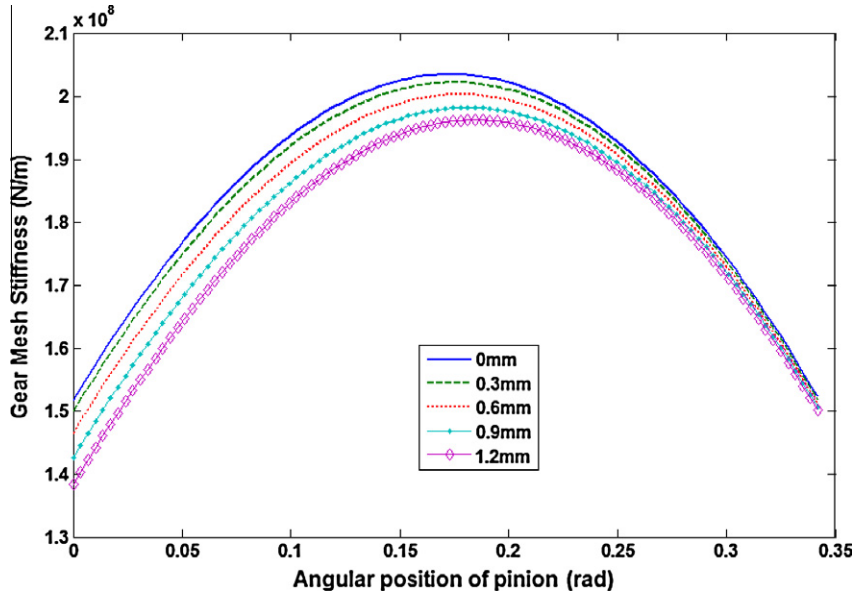


Fig. 8. Mesh stiffness with different crack depth (single tooth pair).

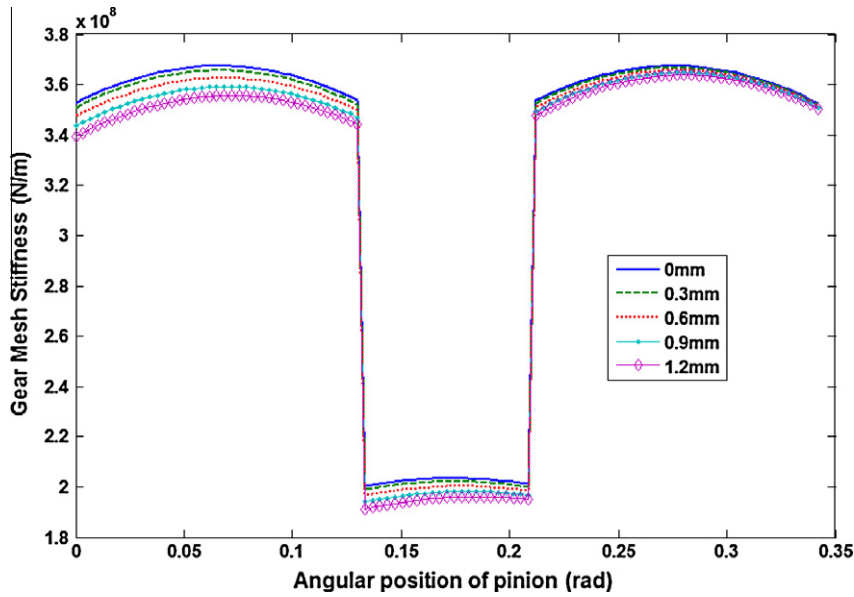


Fig. 9. Mesh stiffness with different crack depth.

K_{ij}/C_{ij} is stiffness/damping of the supporting bearings and $i = p, g$ for pinion and gear, respectively, $j = x, y$ for x and y direction. $k(t)$ is time varying mesh stiffness by which the influence of gear tooth crack is incorporated and C_m is the damping between gear teeth in mesh. F_f is the tooth friction force caused by the sliding between the mating teeth.

The design parameters of the spur gear system applied in this study are shown in Table 4. And this spur gear system operates under a load of 60 nm which is applied to the driven gear. The Coulomb friction model and the dynamic coefficient of friction u measured by Rebbechi et al. [28] are used in this paper.

The equations of motion governing torsional vibration are represented by:

$$J\ddot{\theta} = T \quad (21a)$$

$$J = \begin{bmatrix} J_p & 0 \\ 0 & J_g \end{bmatrix}, \quad \theta = \begin{bmatrix} \theta_p \\ \theta_g \end{bmatrix}, \quad T = \begin{bmatrix} T_p - M_{pN} + M_{pf} \\ -T_g + M_{gN} - M_{gf} \end{bmatrix} \quad (21b)$$

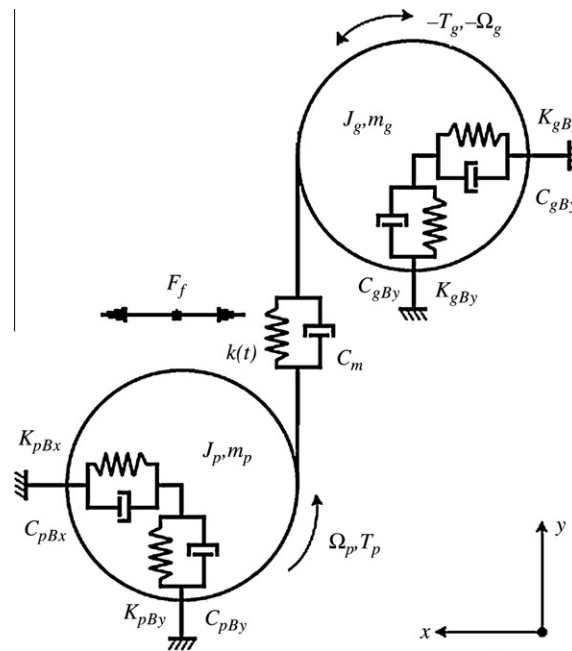


Fig. 10. Scheme of spur gear system with six degrees of freedom.

Table 4
Parameters of the gear system.

	Pinion	Gear
Moment of inertia (kg.m ²)	2×10^{-3}	0.96×10^{-4}
Mass (kg)	0.4439	0.3083
Radial stiffness of the bearing (N/m)	6.56×10^8	6.56×10^8
Damping of the bearing (Ns/m)	1.8×10^3	1.8×10^3
Damping between meshing teeth (Ns/m)		67
Coefficient of friction		0.06

The equations of motion describing the translational vibration are as follows:

$$M\ddot{X} = F \quad (22a)$$

$$M = \begin{bmatrix} m_p & & & \\ & m_g & & \\ & & m_p & \\ & & & m_g \end{bmatrix}, \quad X = \begin{bmatrix} x_p \\ x_g \\ y_p \\ y_g \end{bmatrix} \quad (22b)$$

$$F = \begin{bmatrix} F_f - K_{pBx}x_p - C_{pBx}\dot{x}_p \\ -F_f - K_{gBx}x_g - C_{gBx}\dot{x}_g \\ -N - K_{pBy}y_p - C_{pBy}\dot{y}_p \\ N - K_{gBy}y_g - C_{gBy}\dot{y}_g \end{bmatrix} \quad (22c)$$

Here, N is the net contact force due to the elasticity. $\theta_i/\dot{\theta}_i/\ddot{\theta}_i$ is angular displacement/velocity/acceleration, m_i denotes the mass of gear. $x_i/\dot{x}_i/\ddot{x}_i$ is lateral displacement/velocity/acceleration along x direction. $y_i/\dot{y}_i/\ddot{y}_i$ is lateral displacement/velocity/acceleration along y direction. M_{iN} and M_{if} are the force moments induced, respectively by the normal mesh force along LOA and the corresponding surface friction force based on the method used by He et al. [29]. The supporting bearing stiffness K_{iBj} and damping C_{iBj} are assumed to be constant although some time-varying models have been developed like [30–32] because its beyond the scope of this paper. The subscript $i = p, g$ are for pinion and gear, respectively and $j = x, y$ for x, y directions shown in Fig. 10.

Statistical features which are commonly used to provide a measurement of the vibration level are widely used in mechanical fault detection [33–36]. And Wu et al. [1] studied the performances of some of these statistical indicators when gear

tooth cracks with different size were in presence. A conclusion was drawn that RMS shows the best performance when Method 2 for generating residual signals is used and kurtosis is the most robust indicator no matter what signals are used. The Method 2 is also used in this paper to produce the residual signal and the RMS and kurtosis indicators are also used to predict the severity of the crack propagation. The RMS value is defined as [34]

$$RMS = \sqrt{\frac{1}{N} \sum_{n=1}^N (x(n) - \bar{x})^2}, \quad \bar{x} = \frac{1}{N} \sum_{n=1}^N x(n) \quad (23)$$

And kurtosis is calculated by [37]

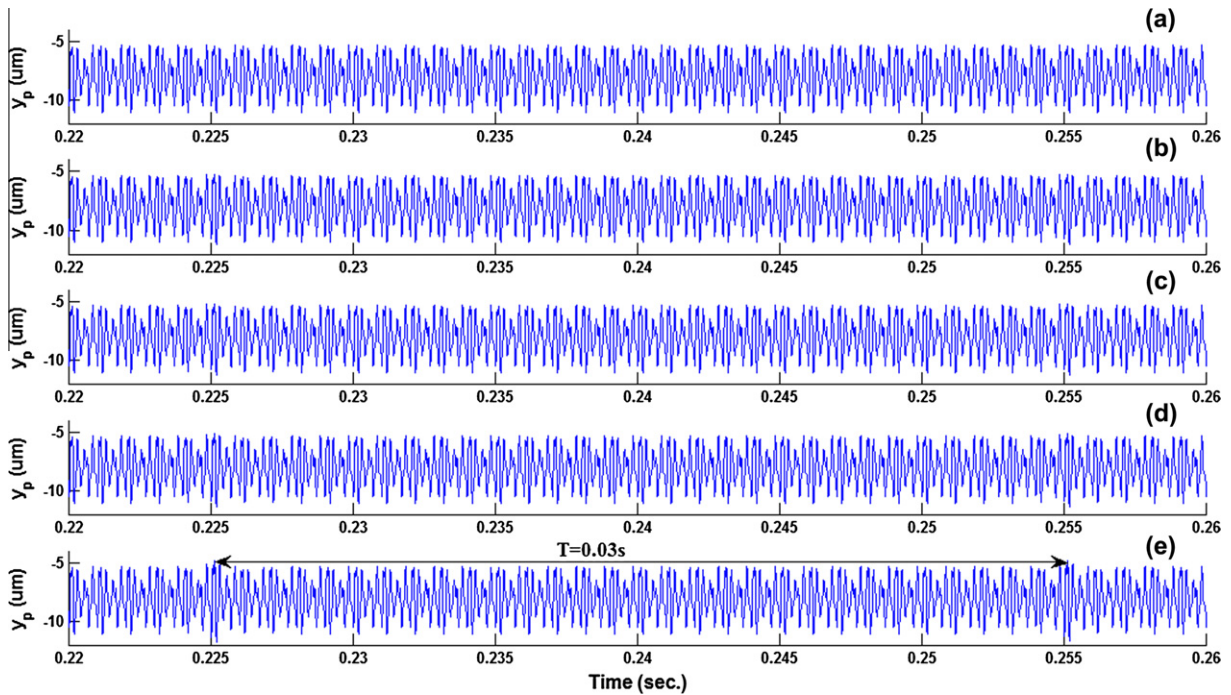


Fig. 11. Displacement of pinion in y direction with crack depth: 1.2 mm, crack inclination angle: $\alpha_c = 60^\circ$ and crack length: (a) 0 mm, (b) 12 mm, (c) 16 mm, (d) 20 mm, and (e) 21.2 mm.

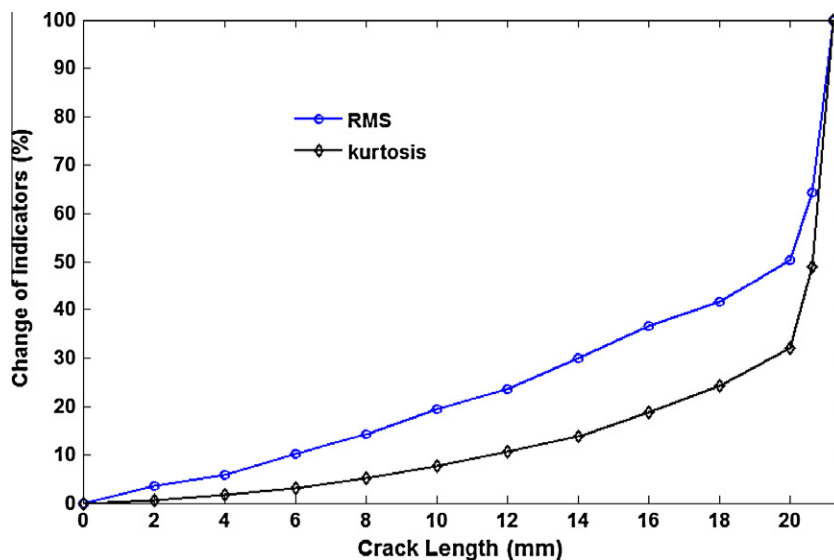


Fig. 12. Change of statistical indicators along crack length.

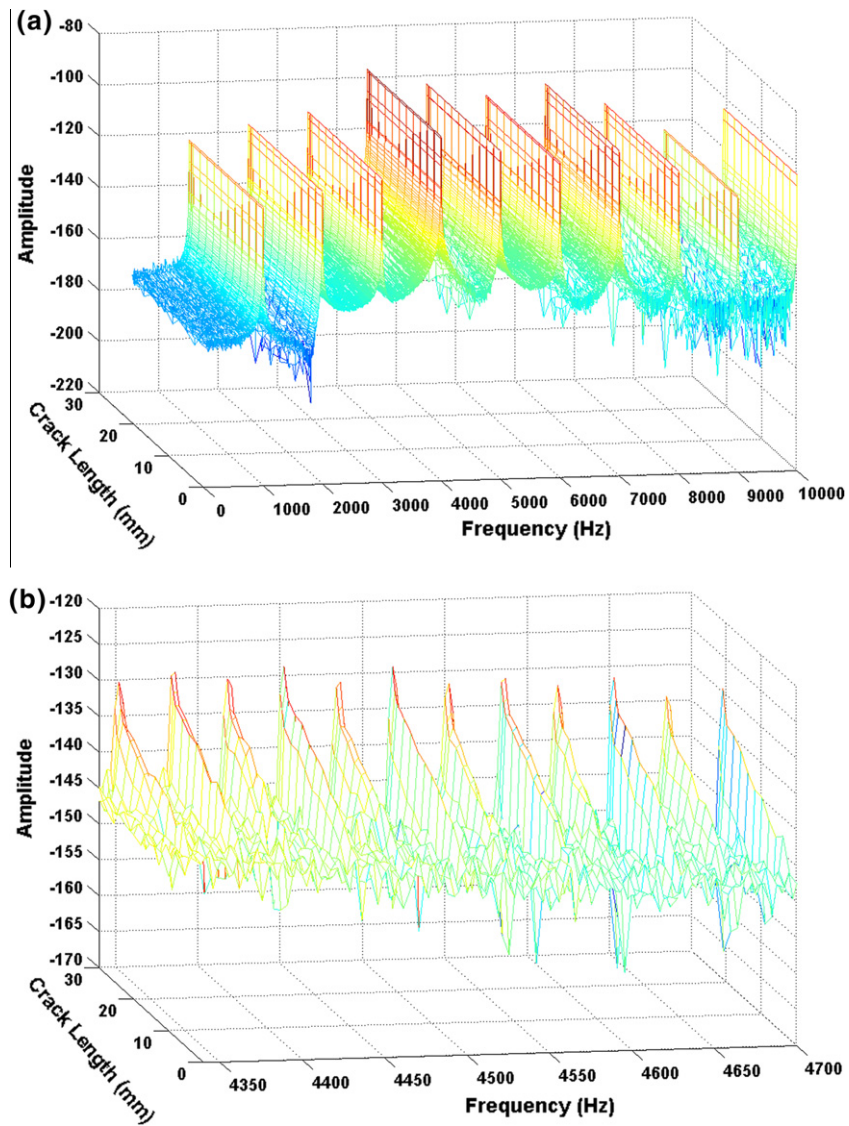


Fig. 13. Spectrum of pinion vibration in y direction with different crack length. (a) Full scope. (b) Zoom plot of (a).

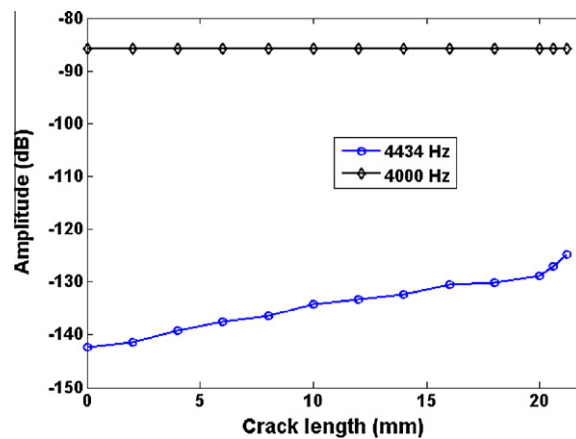


Fig. 14. Amplitude of spectrum of pinion vibration versus crack length.

$$\text{Kurtosis} = \frac{1/N \sum_{n=1}^N (x(n) - \bar{x})^4}{[1/N \sum_{n=1}^N (x(n) - \bar{x})^2]^2} \quad (24)$$

3.2. Dynamic responses of gear system with tooth root crack

The mesh stiffness calculated by the mathematical model discussed afore is incorporated into the dynamic equations of the spur gear system to investigate the effects of gear tooth crack. The simulation is programmed in Matlab with ODE45

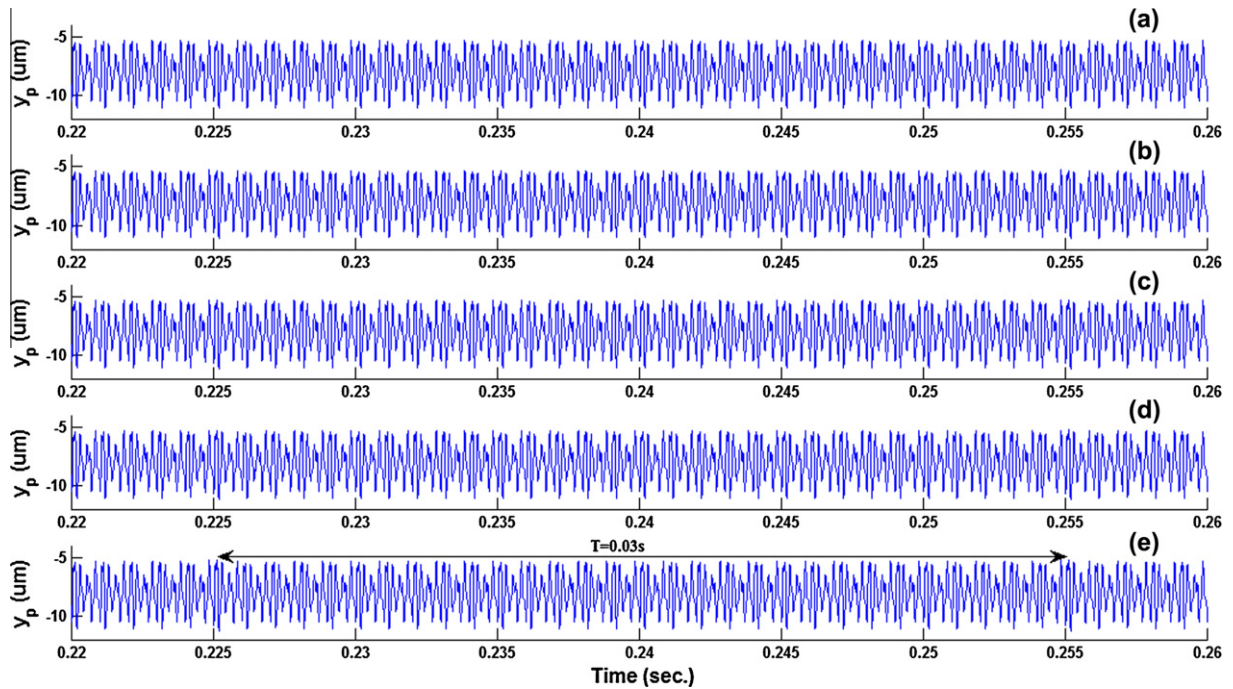


Fig. 15. Displacement of pinion in y direction with crack length: 15 mm, crack inclination angle: $\alpha_c = 60^\circ$ and crack depth: (a) 0 mm, (b) 0.6 mm, (c) 0.8 mm, (d) 1 mm, and (e) 1.2 mm.

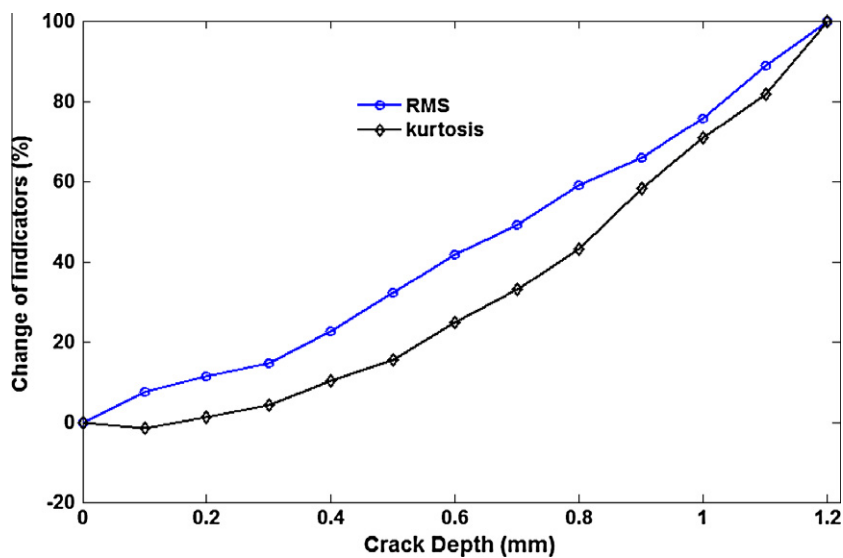


Fig. 16. Change of statistical indicators along crack depth.

subroutine. And the results presented here are exhibited by statistical indicators and observations in time and frequency domain.

The dynamic responses of the spur gear system are simulated. The displacements of the pinion in y direction with different crack length along tooth width in time range of 0.22 and 0.26s under steady state are shown in Fig. 11. It is very hard to observe the change in displacement waveform produced by the gear tooth root crack when it is at early stage, such as the Fig. 11b and c. However, two distinct impulses in Fig. 11d and e are in presence when the crack propagates to a certain degree. And the time interval between the two adjacent impulses are exactly equal to the rotational period of the pinion, that is $60/2000 = 0.03$ s.

The statistical indicators – RMS and kurtosis defined by formulas (23) and (24), respectively are used to assess the severity of tooth root crack and the curves showing the trends of these two indicators along tooth crack length are displayed in Fig. 12. Here, the largest value corresponding to the case with largest tooth crack length, that is $W_c = W$, $q_2 = q_0$, minus that of the healthy tooth is defined as 100%. It is noticed that the tooth width is 20 mm and the crack length greater than tooth width in Fig. 12 means that the crack has gone through the whole tooth width and been propagating along crack depth q_2 which is described in Fig. 4. When the tooth crack propagates along tooth width, both RMS and kurtosis appear to be linear with the crack length. However, there is a prompt increase for the two indicators when the crack length exceeds the tooth width. Change the statistical indicators due to the tooth crack propagation can be useful for gear fault diagnosis and condition monitoring.

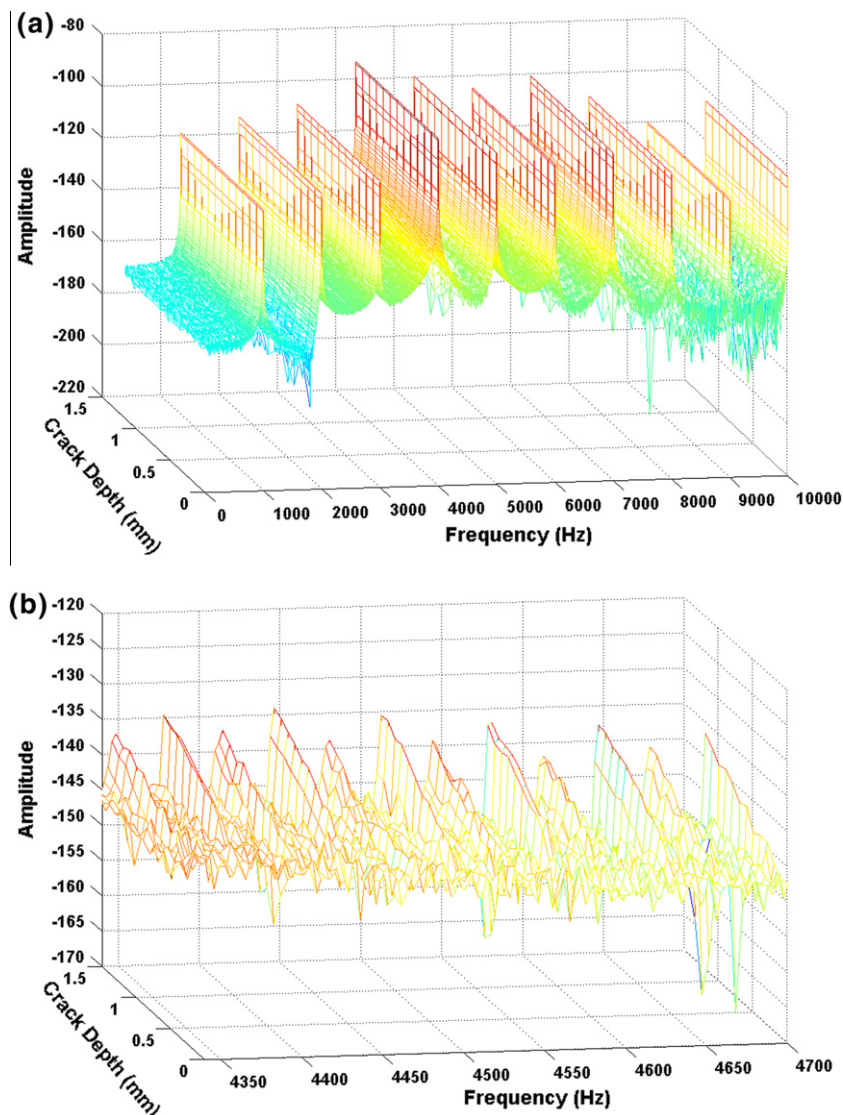


Fig. 17. Spectrum of pinion vibration in y direction with different crack depth. (a) Full scope. (b) Zoom plot of (a).

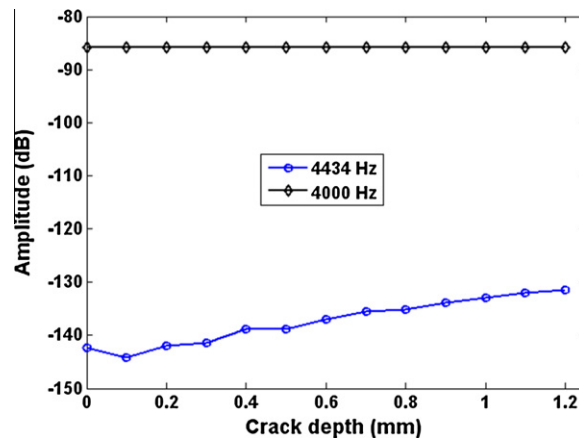


Fig. 18. Amplitude of spectrum of pinion vibration versus crack depth.

It is well known that, the tooth-meshing frequency and its harmonics, sometimes together with sideband structures due to modulation effects, are the most important components in gear vibration spectra [38]. The increment in the number and amplitude of sidebands may indicate a gear fault condition, and the spacing of the side-bands is related to their source [39]. Consequently, structures of sideband can be used as a useful diagnostic feature for gear fault detection. As the same as observed in published papers like Refs. [13,31,32,38–40], this paper will show the similar phenomenon and obtain some useful results. Influence of gear tooth crack propagating along tooth width on the frequency characteristics is investigated and shown in Figs. 13 and 14. In Fig. 13, the plot (a) shows the spectrum of the dynamic response of the pinion in y direction while (b) is the zoomed plot exhibiting how the tooth crack changes the sidebands around the mesh frequency and its harmonics. And the changes in the amplitudes of the fourth harmonic component and one of its sideband are shown in Fig. 14 as the growth of tooth crack length. Consequently, a conclusion can be drawn that the amplitudes of the spectrum near mesh frequency and its harmonics are hardly changed by the gear tooth crack. By contrast, the sidebands begin to appear when the tooth crack appears and their magnitudes increase with the crack propagation which can be observed clearly in plot (b). It means that the sidebands are more sensitive to gear tooth crack propagation than the mesh frequency and its harmonics components.

The same analysis are also carried out to study the changes of the dynamic response characteristics due to gear tooth crack propagation along crack depth while keeping the crack length and inclination angle unchanged ($W_c = 15$ mm and $\alpha_c = 60^\circ$). And the corresponding results are shown in Figs. 15–18. The displacements of pinion in y direction with different crack depth are shown in Fig. 15. The statistical indicators – RMS and kurtosis defined by formulas (23) and (24), respectively are also applied to explore the effect of the crack propagation along depth and shown in Fig. 16. As defined before, the 100% is equal to the indicators' largest value corresponding to the largest tooth crack depth ($W_c = 15$ mm and $q_0 = 1.2$ mm) minus that of the healthy tooth. When the tooth crack propagates along tooth depth while keeping the crack length and inclination angle constant values ($W_c = 15$ mm and $\alpha_c = 60^\circ$), both RMS and kurtosis appear to be near a quadratic versus the crack length.

Influence of gear tooth crack propagating along tooth depth on the frequency spectrum characteristics shown in Figs. 17 and 18 is also performed. In Fig. 17, the spectrum of the dynamic response of the pinion in y direction is plotted in (a) while plot (b) is the zoom plot which gives the information about how the tooth crack influences the sidebands around the mesh frequency and its harmonics. The fourth mesh harmonic component and one of its sideband changing with the tooth crack depth are shown in Fig. 18. As suggested in Figs. 17 and 18, it can be concluded that the amplitudes at the frequencies near mesh frequency and its harmonics are nearly not affected by the gear tooth crack. However, the sidebands around the mesh frequency and its harmonics begin to appear with the presence of the tooth crack and their magnitudes increase with the crack propagation which can be observed clearly in plot (b). Likewise, it can be concluded that the sidebands are more sensitive to gear tooth crack than the mesh frequency and its harmonics. Observation in the change of the sidebands can offer more information on the presence and severity of tooth crack.

4. Conclusions

An analytical mesh stiffness model of spur gear with tooth root crack propagating along both tooth width and crack depth is proposed in this paper. It is validated by comparison with the FEA results. At the same time, effects of gear tooth crack propagating along tooth width and depth on gear mesh stiffness are investigated quantitatively. A dynamic model of spur gear pair system having six-degree-of-freedom is developed to examine the influence of gear tooth crack by incorporation of the developed mesh stiffness model. And two statistical indicators – RMS and kurtosis are used to assess the severity of

the gear tooth crack and how the tooth crack changes the frequency structure of the gear dynamic response. A conclusion can be drawn that both the RMS and kurtosis increase with the crack propagation whatever along tooth width or crack depth. The gear tooth crack can also result in the presence of the sidebands around mesh frequency and its harmonics, and their magnitudes increase with tooth crack propagation along both tooth width and crack depth. The proposed model in this paper is limited to spur gear pairs and to develop analytical models of gear mesh stiffness with different type of gear tooth faults which will be suitable for different kind of gear pairs, is the scope of our future research work.

Acknowledgment

The authors are grateful for the financial support provided by the National Natural Science Key Foundation of China under Contract No. 51035008.

References

- [1] Wu S, Zuo MJ, Parey A. Simulation of spur gear dynamics and estimation of fault growth. *J Sound Vib* 2008;317:608–24.
- [2] Loutridis SJ. Instantaneous energy density as a feature for gear fault detection. *Mech Syst Signal Process* 2006;20(5):1239–53.
- [3] Chaari F, Baccar W, Abbas MS, et al. Effect of spalling or tooth breakage on gearmesh stiffness and dynamic response of a one-stage spur gear transmission. *Eur J Mech A – Solids* 2008;27:691–705.
- [4] Chaari F, Fakhfakh T, Haddar M. Analytical modelling of spur gear tooth crack and influence on gearmesh stiffness. *Eur J Mech A – Solids* 2009;28:461–8.
- [5] Sirichai S. Torsional properties of spur gears in mesh using nonlinear finite element analysis, PhD. Curtin University of Technology; 1999.
- [6] Pimsarn M, Kazerounian K. Efficient evaluation of spur gear tooth mesh load using pseudo-interference stiffness estimation method. *Mech Mach Theory* 2002;37:769–86.
- [7] Wang J. Numerical and experimental analysis of spur gears in mesh, PhD thesis. Curtin University of Technology; 2003.
- [8] Shao YM, Chen ZG, Wang SM. Simulation of spur gear pair with faults using FEA. In: ICROS-SICE international joint conference 2009; August 18–21, 2009 [Japan].
- [9] Weber C. The deformation of loaded gears and the effect on their load carrying capacity, Sponsored research (Germany). British Dept of Sci and Ind Res Report No. 3; 1949.
- [10] Cornell RW. Compliance and stress sensitivity of spur gear teeth. *ASME J Mech Des* 1981;103:447–59.
- [11] Kasuba R, Evans JW. An extended model for determining dynamic loads in spur gearing. *J Mech Des Trans Am Soc Mech Eng* 1981;103(2):398–409.
- [12] Yang DCH, Lin JY. Hertzian damping, tooth friction and bending elasticity in gear impact dynamics. *J Mech Transm Aut Des* 1987;109(2):189–96.
- [13] Tian XH. Dynamic simulation for system response of gearbox including localized gear faults. Master's thesis. Edmonton, Alberta, Canada: University of Alberta; 2004.
- [14] Lewicki DG. Effect of rim thickness on gear crack propagation path. NASA/TM-1996-107229, ARL-TR-1110.
- [15] Lewicki DG. Gear crack propagation investigations. NASA/TM-1996-107147, ARL-TR-597.
- [16] Lewicki DG. Gear crack propagation path studies-guidelines for ultra-safe design. NASA/TM-2001-211073, ARL-TR-2468.
- [17] Özgüven HN, Houser DR. Mathematical models used in gear dynamics – a review. *J Sound Vib* 1988;121(3):383–411.
- [18] Parey A, Tandon N. Spur gear dynamic models including defects – a review. *Shock Vib Dig* 2003;35(6):465–78.
- [19] Randall RB. The application of fault simulation to machine diagnostics and prognostics. *Int J Acoust Vib* 2009;14(2):81–9.
- [20] Randall RB. The use of simulation models to generate data corresponding to faults in machines. In: Proceedings of 2009 8th international conference on reliability, Maintainability and safety. ICRMS; 2009. p. 934–39.
- [21] Li CJ, Lee H, Choi SH. Estimating size of gear tooth root crack using embedded modeling. *Mech Syst Signal Process* 2002;16(5):841–52.
- [22] Li CJ, Lee H. Gear fatigue crack prognosis using embedded–dynamic–fracture model, gear dynamic model and fracture mechanics. *Mech Syst Signal Process* 2005;19:836–46.
- [23] Endo H, Randall RB, Gosselin C. Differential diagnosis of spall versus cracks in the gear tooth fillet region. *JFAPBC* 2004;5:63–71.
- [24] Endo H, Randall RB, Gosselin C. Differential diagnosis of spall versus cracks in the gear tooth fillet region: experimental validation. *Mech Syst Signal Process* 2009;23:636–51.
- [25] Yang DCH, Su ZS. A rotary model for spur gear dynamics. *ASME J Mech Transm Aut Des* 1985;107:529–35.
- [26] Sainsot P, Velex P, Duverger O. Contribution of gear body to tooth deflections – a new bidimensional analytical formula. *ASME J Mech Des* 2004;126:748–52.
- [27] Muskhelishvili NL. Some basic problems of the mathematical theory of elasticity. 2nd ed. English ed. The Netherlands: P. Noordhoff Limited; 1975.
- [28] Rebbechi B, Oswald F, Townsend D. Measurement of gear tooth dynamic friction. *ASME, DE – vol. 88*. In: Proceedings of the 7th power transmission and gearing conference; 1996. p. 355–63.
- [29] He S, Cho S, Singh R. Prediction of dynamic friction forces in spur gears using alternate sliding friction formulations. *J Sound Vib* 2008;309:843–51.
- [30] Viadero F, Fernandez del Rincon A, Sancibrian R, et al. A model of spur gears supported by ball bearings. *WIT transactions on modelling and simulation*. vol 46, © WIT Press; 2007. p. 711–22.
- [31] Sawalhi N, Randall RB. Simulating gear, bearing interactions in the presence of faults. Part I The combined gear bearing dynamic model and the simulation of localised bearing faults. *Mech Syst Signal Process* 2008;22:1924–51.
- [32] Sawalhi N, Randall RB. Simulating gear, bearing interactions in the presence of faults. Part II Simulation of the vibrations produced by extended bearing faults. *Mech Syst Signal Process* 2008;22:1952–66.
- [33] Lebold M, McClintic K, Campbell R, et al. Review of vibration analysis methods for gearbox diagnostics and prognostics. Proceedings of the 54th meeting of the society for machinery failure prevention technology. Virginia Beach, VA; May 1–4, 2000. p. 623–34.
- [34] Forrester BD. Advanced vibration analysis techniques for fault detection and diagnosis in geared transmission systems, PhD thesis. Swinburne University of Technology; 1996.
- [35] Honarvar F, Marin HR. New statistical moments for diagnostics of rolling element bearings. *J Manuf Sci Eng* 1997;119:425–32.
- [36] Stewart RM. Some useful data analysis techniques for gearbox diagnostics, applications of time series analysis, PhD thesis. ISVR, University of Southampton; 1977.
- [37] Dron JP, Bollaers F, Rasolofondraibe I. Improvement of the sensitivity of the scalar indicators (crest factor, kurtosis) using a denoising method by spectral subtraction: application to the detection of defects in ball bearings. *J Sound Vib* 2004;270(1–2):61–73.
- [38] Khang NV, Cau TM, Dien NP. Modelling parametric vibration of gear-pair systems as a tool for aiding gear fault diagnosis. *Technische Mechanik Band* 24 Heft 2004;3–4:198–205.
- [39] Dalpiaz G, Rivola A, Rubini R. Effectiveness and sensitivity of vibration processing techniques for local fault detection in gears. *Mech Syst Signal Process* 2000;14(3):387–412.
- [40] Jia S, Howard I. Comparison of localised spalling and crack damage from dynamic modelling of spur gear vibrations. *Mech Syst Signal Process* 2006;20:332–49.